

VOCATIONAL TRAINING REPORT

Operational Analysis and Case Study of Amalgamated Keshalpur–West Mudidih Colliery,
Katras Area, B.C.C.L.

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Training Period: **28 Days – May 2025**
Organization: Bharat Coking Coal Limited (BCCL)
Location: Katras Area, Dhanbad, Jharkhand

ACKNOWLEDGEMENT

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INTRODUCTION

Bharat Coking Coal Limited (BCCL), a subsidiary of Coal India Limited (CIL), operates coal mines in the Jharia Coalfield. Katras Area is an operational area of BCCL in Dhanbad. The training provided practical exposure to underground and opencast mining, ventilation, safety, drilling and blasting, transportation, drainage and mechanized excavation.

CASE STUDY: KATRAS AREA OPERATIONS

The principal mine studied was the **Amalgamated Keshalpur–West Mudidih Colliery**. It has mixed underground and opencast workings and multiple coal seams. Training exposure included underground visits, ventilation direction, methane monitoring, coal transportation, drainage, and surface drilling, blasting, excavation and shovel–dumper operations.

OBJECTIVES OF TRAINING

- To study underground and opencast coal-mining operations.
- To understand the geology and occurrence of coal seams.
- To observe ventilation, methane monitoring and mine safety.
- To understand drilling, blasting, excavation and transportation.
- To correlate classroom knowledge with field practice.

MINE DETAILS

Mine Name: Amalgamated Keshalpur–West Mudidih Colliery

Area: Katras Area, BCCL

Type: Mixed underground and opencast coal mine

Coalfield: Jharia Coalfield

Method: Bord-and-Pillar (underground) and mechanized opencast working

Coal Seams: 11 seams reported in the training material

Underground / Opencast: Seams 1–2 / Seams 3–11

Thickest Seam: 4th seam, approximately 15–16 m

Major Fault: Normal fault with approximately 71 m throw

Mine Area / Depth: Approximately 5.8 km²; workings reported to about 150 m depth

Excavation: Shovel–dumper system

Drilling: Two drill machines reported in the training material

OPERATIONAL ANALYSIS

1. Underground Mining and Ventilation: The underground visit provided exposure to bord-and-pillar workings, mine entry, ventilation arrangements and airflow direction. Ventilation is essential for supplying fresh air, diluting gases and maintaining safe working conditions.

2. Methane Monitoring and Safety: Methane monitoring was studied as an important underground safety requirement. Proper ventilation, gas monitoring, PPE and compliance with mine procedures are essential for safe coal mining.

3. Transportation and Drainage: The training included exposure to belt conveyors and haulage used for coal/material transportation. Mine drainage and pumping arrangements help prevent water accumulation and maintain safe working conditions.

4. Drilling and Blasting: In opencast operations, drilling creates blast holes for explosive charging. Blasting fragments rock/overburden for subsequent excavation. Burden, spacing, stemming, fragmentation and blast safety are important considerations.

5. Loading and Transportation: The shovel–dumper combination forms an important part of the opencast production cycle. Shovels excavate and load material into dumpers, which transport it to the designated destination. Proper equipment coordination improves productivity.

6. Geological Considerations: Multiple seams, the thick 4th seam and the major normal fault influence mine planning and extraction. Geological structures must be considered while developing and extending mine workings.

LEARNING OUTCOMES

- Understood underground and opencast coal-mining operations.
- Studied bord-and-pillar mining and mine ventilation.
- Gained exposure to methane monitoring, transportation and mine drainage.
- Learned the relationship between drilling, blasting, excavation and haulage.
- Gained practical understanding of mine safety and field operations.

CONCLUSION

The 28-day vocational training at Katras Area, BCCL, provided valuable industrial exposure to underground and opencast coal-mining operations. It helped me understand the practical application of geology, bord-and-pillar mining, ventilation, methane monitoring, transportation, drainage, drilling, blasting and shovel–dumper operations. Overall, the training strengthened my practical Mining Engineering knowledge and bridged the gap between classroom theory and actual mine practice.